Project Report

Global Fuel Prices and Taxation Trends

Data Analytics & Data Visualization using Tableau Public

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Project Name	Global Fuel Prices and Taxation Trends
Tool	Tableau Public
Platform	SkillWallet
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1. INTRODUCTION

1.1 Project Overview

This project analyses global fuel prices and fuel taxation using Tableau Public. The dataset contains records for different countries, with details such as country, date, income level, region, subsidy level, Brent crude oil price, and the price per litre of diesel, LPG and petrol, along with the fuel tax percentage.

The raw data is converted into nine visualizations that compare prices by country, region and income level, show price trends over time, display petrol prices on a world map, and explore the relationship between fuel tax and petrol price. These are combined into an interactive dashboard and a data story, and published on Tableau Public so that anyone can explore the results through a web link.

1.2 Purpose

The purpose of the project is to make fuel price and tax data easy to understand through visual analysis. The project aims to:

- Compare petrol and diesel prices across countries, regions and income groups.
- Show how fuel prices have changed over time.
- Study how fuel tax percentages relate to retail fuel prices.
- Provide an interactive dashboard that lets users filter the data and find insights themselves.
- Practise the complete data analytics workflow: data collection, preparation, visualization, dashboard, story, testing and web publishing.

2. IDEATION PHASE

2.1 Problem Statement

Fuel prices differ widely from one country to another, and the reasons include taxes, subsidies, income level and crude oil prices. The data is usually available only as large tables, which makes it hard for students, analysts and general readers to compare countries or notice patterns. There is a need for a clear, interactive visual tool that explains how global fuel prices and taxation vary.

Problem Statement	I am	I'm trying to	But	Because	Which makes me feel
PS-1	A data analyst / student	Compare fuel prices and taxes across countries	I cannot see patterns in raw tables	The dataset is large and has many columns	Confused and slow in finding insights

2.2 Empathy Map Canvas

The empathy map below is for the main user of the dashboard: a student or analyst who wants to understand fuel prices and taxes.

Says	Thinks
"Why is petrol so costly in some countries?" "Which region has the lowest fuel prices?"	Taxes might be the main reason for high prices. A chart would be easier to read than a spreadsheet.

Does	Feels
Searches for fuel price data online, compares a few countries manually, and looks for charts and maps.	Overwhelmed by large tables; interested when data is shown visually; wants quick and trustworthy answers.

2.3 Brainstorming

Several ideas were discussed for presenting the data. The ideas and the final decision are listed below.

Idea	Description	Decision
Bar charts for countries	Compare petrol, diesel and tax by country using sorted bars	Selected
Line chart for trends	Show monthly petrol price movement for a few countries	Selected
World map	Colour countries by average petrol price	Selected
Scatter plot	Show relationship between tax percentage and petrol price	Selected
Region and income level charts	Compare average prices between groups	Selected
Interactive dashboard and story	Combine all charts with filters and a guided narrative	Selected
Price prediction model	Forecast future prices	Kept for future scope

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Awareness	Exploration	Analysis	Insight
User action	Opens the dashboard link	Uses filters for country, region and income level	Compares charts, map and scatter plot	Reads the story points
User thinking	"What does this show?"	"Where is my country?"	"Why do prices differ?"	"Tax and income level matter"
Pain point	Too much raw data elsewhere	Needs quick filtering	Needs fair comparison (average, not sum)	Wants a simple summary
Opportunity	Clear title and layout	Interactive filters	Consistent average values	Short story with key findings

3.2 Solution Requirement

Functional requirements

- Connect the fuel price dataset to Tableau and check data types and missing values.
- Build nine worksheets using average values for fair comparison.
- Combine the worksheets into one dashboard with filters and actions.
- Create a data story that explains the main insights.
- Publish the dashboard to Tableau Public and share the link.

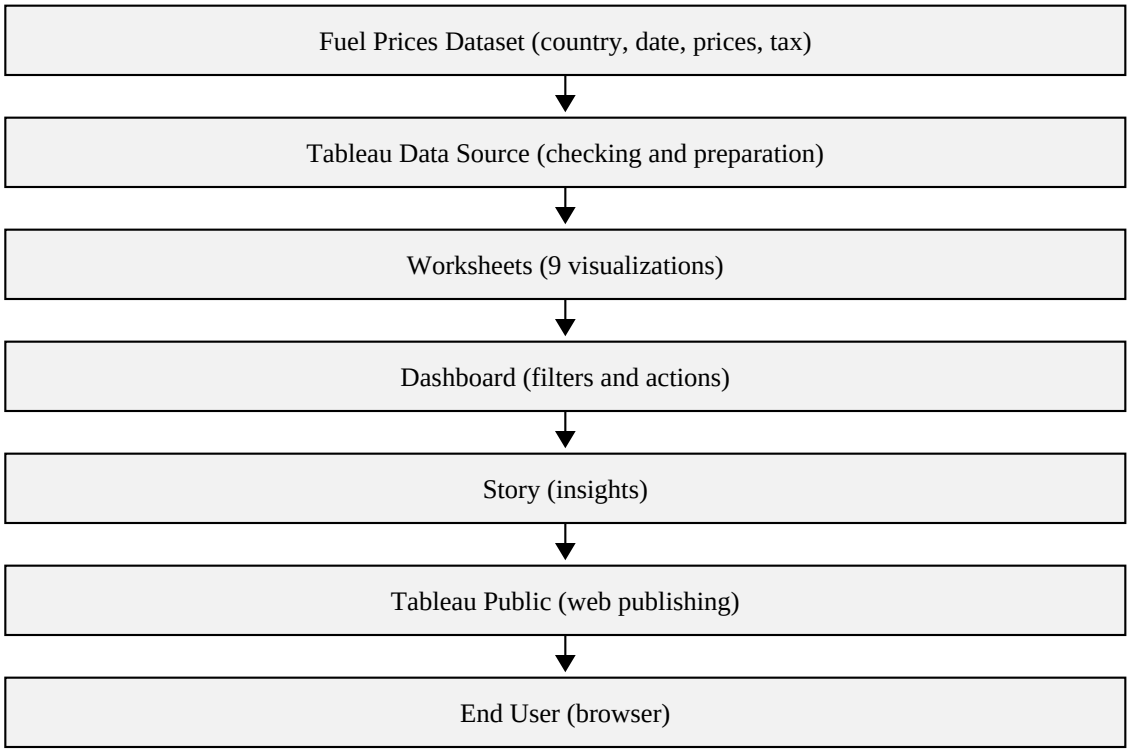
Non-functional requirements

- Dashboard should load quickly and respond smoothly to filters.

- Charts should be readable, with clear titles, labels and colours.
- Data should be accurate and consistent (for example, tax percentage format).
- Dashboard should be accessible through a web browser without installation.

3.3 Data Flow Diagram

The diagram shows how data moves from the source dataset to the final user.



3.4 Technology Stack

Component	Technology / Tool	Use
Data	Global fuel prices dataset (CSV / Excel)	Source of prices, tax and country details
Analytics & Visualization	Tableau Public	Data preparation, worksheets, dashboard, story
Platform	SkillWallet	Project guidance and submission
Publishing	Tableau Public web	Hosting the interactive dashboard
Access	Web browser	Viewing the dashboard on any device

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem	Solution
Fuel data is hard to understand in table form	Visual charts, map and scatter plot make patterns visible
Sum of values gives misleading comparison	Average (AVG) aggregation is used for all price comparisons

Problem	Solution
Too many countries make charts cluttered	Filters and focused selections (3 to 5 countries) for trends
Users want to explore on their own	Interactive dashboard with filters on country, region and income level
Results need to be easily shared	Published on Tableau Public with a web link

4.2 Proposed Solution

The proposed solution is an interactive Tableau dashboard for global fuel prices and taxation. After connecting and checking the dataset, nine worksheets are created: average crude oil price by country, petrol price by country, petrol vs diesel comparison, fuel price trend over time, average fuel prices by region, tax percentage by country, fuel prices by income level, global petrol price map, and fuel price vs tax relationship. The worksheets are arranged in a dashboard with filters and then explained through a story, and the final work is published on Tableau Public.

Key features:

- Nine different visualization types in one project.
- Interactive filters and highlighting across charts.
- World map with tax details in the tooltip.
- Scatter plot with trend line for price and tax.
- Story that guides the viewer through the insights.

4.3 Solution Architecture

The solution follows a simple layered architecture, from data to the user.

Layer	Description
Data layer	Fuel price dataset with country, date, income level, region, subsidy level, Brent crude price, diesel, LPG and petrol price per litre, and tax percentage
Preparation layer	Tableau Data Source tab: data type check, missing value check, tax format check
Analysis layer	Calculations using average aggregation; filters, sorting, colour and trend lines
Visualization layer	Nine worksheets: bar charts, line chart, map, scatter plot
Presentation layer	Dashboard and story
Delivery layer	Tableau Public web link opened in a browser

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

S.No	Phase	Activities	Status
1	Data Collection & Extraction	Collect the fuel price dataset and connect it to Tableau	Completed
2	Data Preparation	Check data types, missing values and tax format	Completed

S.No	Phase	Activities	Status
3	Data Visualizations	Create nine worksheets with average aggregation	Completed
4	Dashboard	Combine worksheets with filters and actions	Completed
5	Story	Create story points with insights	Completed
6	Performance Testing	Test load time, filters and calculations	Completed
7	Web Integration	Publish to Tableau Public and share link	Completed
8	Project Demonstration & Documentation	Prepare demo plan and project report	Completed

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

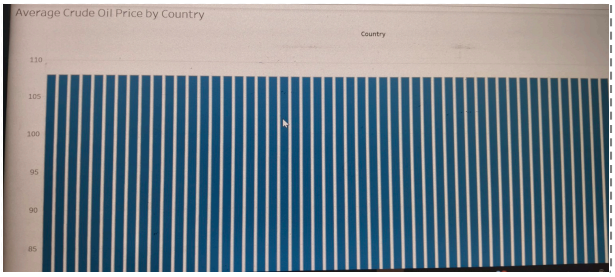
The dashboard was tested for correctness, speed and usability before publishing. The results are summarised below.

S.No	Test	Expected Result	Result
1	Data check	Prices are numbers and missing values are identified	Pass
2	Aggregation check	Average is used instead of sum for prices	Pass
3	Tax format check	Tax percentage shown in one consistent format	Pass
4	Filter test	Country, region and income filters update the charts	Pass
5	Map test	Countries are recognised on the map and show tooltip details	Pass
6	Load and response time	Dashboard opens and responds quickly	Pass
7	Link access	Published Tableau Public link opens in a browser	Pass

7. RESULTS

7.1 Output Screenshots

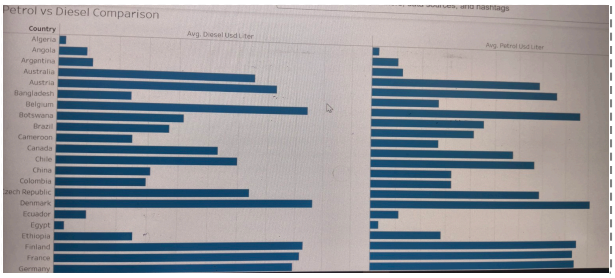
The output screenshots of the worksheets, dashboard and story are shown below.



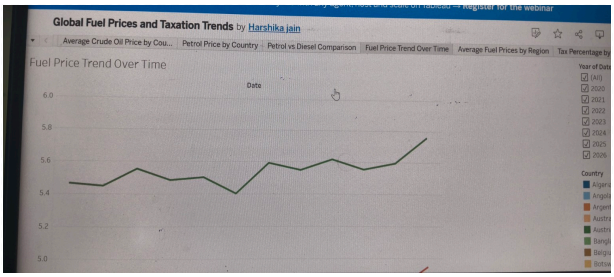
Average Crude Oil Price by Country



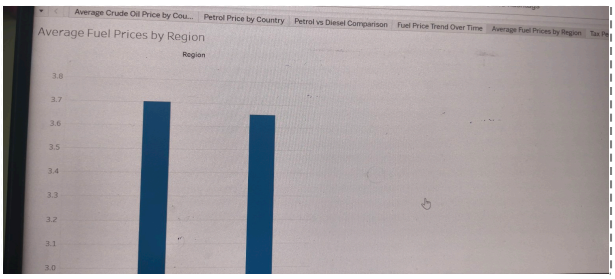
Petrol Price by Country



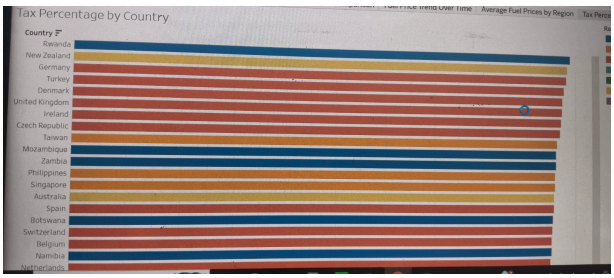
Petrol vs Diesel Comparison



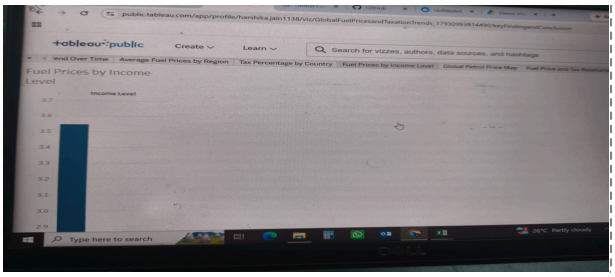
Fuel Price Trend Over Time



Average Fuel Prices by Region



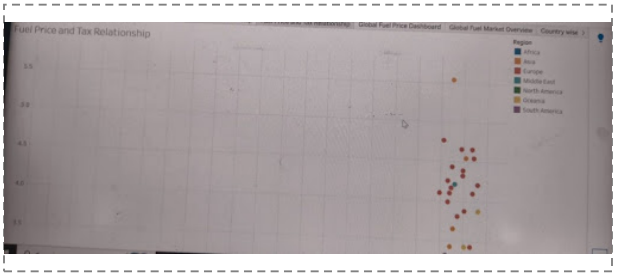
Tax Percentage by Country



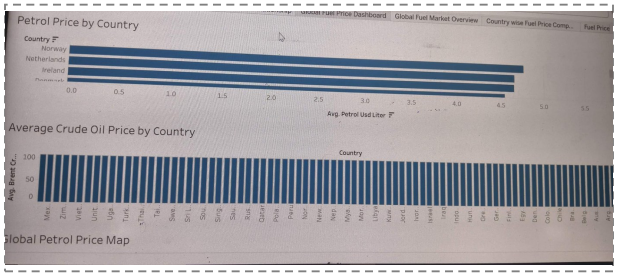
Fuel Prices by Income Level



Global Petrol Price Map



Fuel Price vs Tax Relationship



Dashboard



Story

8. ADVANTAGES & DISADVANTAGES

Advantages

- Easy to understand: charts, map and scatter plot explain the data better than tables.
- Interactive: users can filter and explore the data on their own.
- Fair comparison: average values are used instead of sums.
- Accessible: available online through a link, with no software installation.
- Covers the full workflow from raw data to a published dashboard.

Disadvantages

- Data is loaded manually and is not updated in real time.
- Tableau Public makes the dashboard and data public and has data size limits.
- Only fuel prices and tax are included, so the reasons for price differences cannot be fully explained.
- Results depend on the quality and coverage of the dataset.

9. CONCLUSION

The project converts a large fuel price and tax dataset into a clear, interactive Tableau dashboard. By using average values, consistent formatting and nine different visualizations, it allows users to compare countries, regions and income levels, follow price changes over time, and study how fuel tax relates to petrol prices. The dashboard and story, published on Tableau Public, show how data visualization can make complex global data simple to explore and understand.

10. FUTURE SCOPE

- Connect to live or regularly updated fuel price sources for automatic refresh.
- Add inflation, exchange rate, income and subsidy data to explain price differences better.
- Add price forecasting to estimate future fuel prices.
- Make a mobile-friendly dashboard layout.
- Move to Tableau Cloud or Server for larger audiences and controlled access.

11. APPENDIX

Item	Details
Source Code (if any)	Not applicable. The project is built in Tableau Public using drag-and-drop worksheets, so there is no source code.
Dataset Link	https://huggingface.co/datasets/EduDevCommons/Global-Fuel-Prices
GitHub & Project Demo Link	GitHub: https://github.com/Harshikajain2006/Global-Fuel-Taxation-Trends Tableau Public dashboard: https://public.tableau.com/app/profile/harshika.jain1138/viz/GlobalFuelPriceDashboard/GlobalFuelPriceDashboard